DATAWISE AI

DataWise AI: End-User Guide

Interactive Manual for Automated Data Analysis, Modeling, and Artifact Delivery

Document Type:	End-User Practical Handbook
Target Audience:	Data Scientists, Business Analysts, Product Managers
Document Version:	v1.0.0 (Engineering Release)
Date of Issue:	October 27, 2026
Repository Status:	Verified Codebase & Production Architecture
Orchestration:	LangGraph 16-Agent Autonomous Architecture

Architectural Foundation: Deterministic ML execution engine paired with autonomous multi-agent reasoning, strict data leakage boundaries, and human-in-the-loop validation.

1. Getting Started with DataWise AI

Welcome to DataWise AI. This guide walks you through uploading your dataset, communicating your business objectives, reviewing autonomous agent recommendations, customizing machine learning models, and exporting production-ready code.

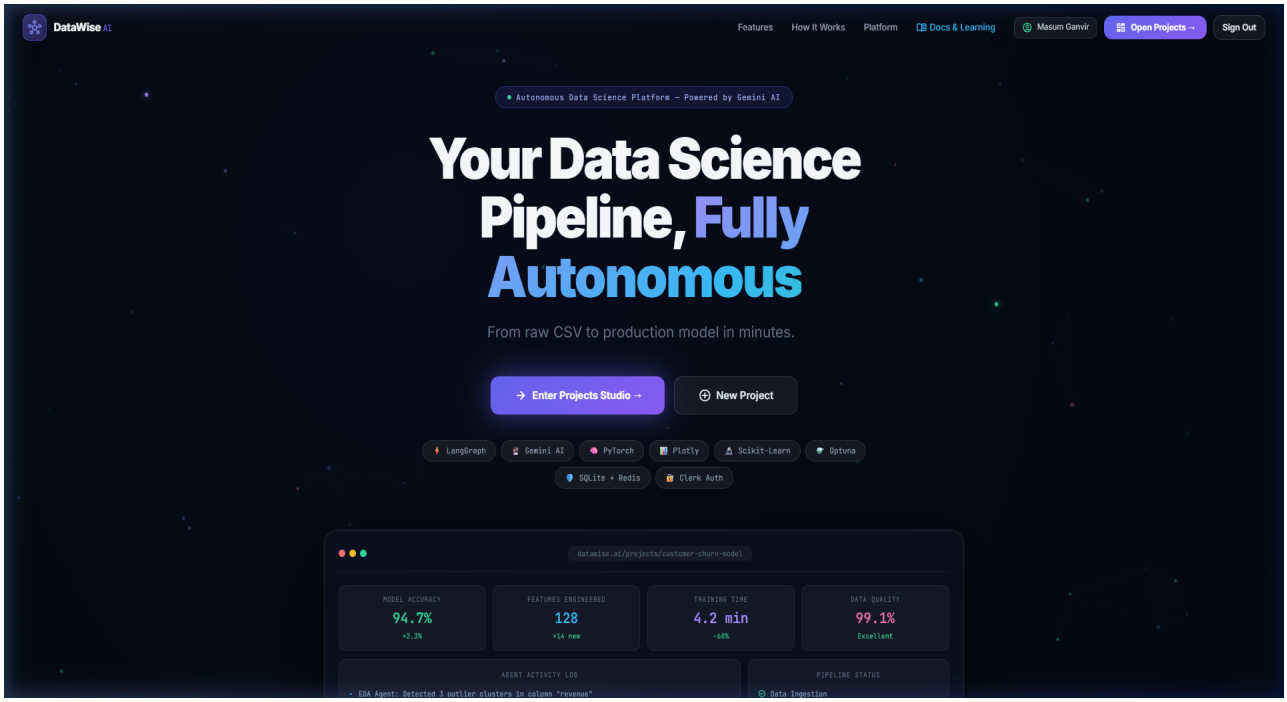


Figure: Step 1: Accessing DataWise AI via Secure Authentication Gateway

2. Creating a New Project & Uploading Data

From the Dashboard, click **'New Project'**. Upload your tabular dataset in CSV, Excel (.xlsx, .xls), or JSON format. The system immediately conducts an automated security scan and presents an interactive data preview.

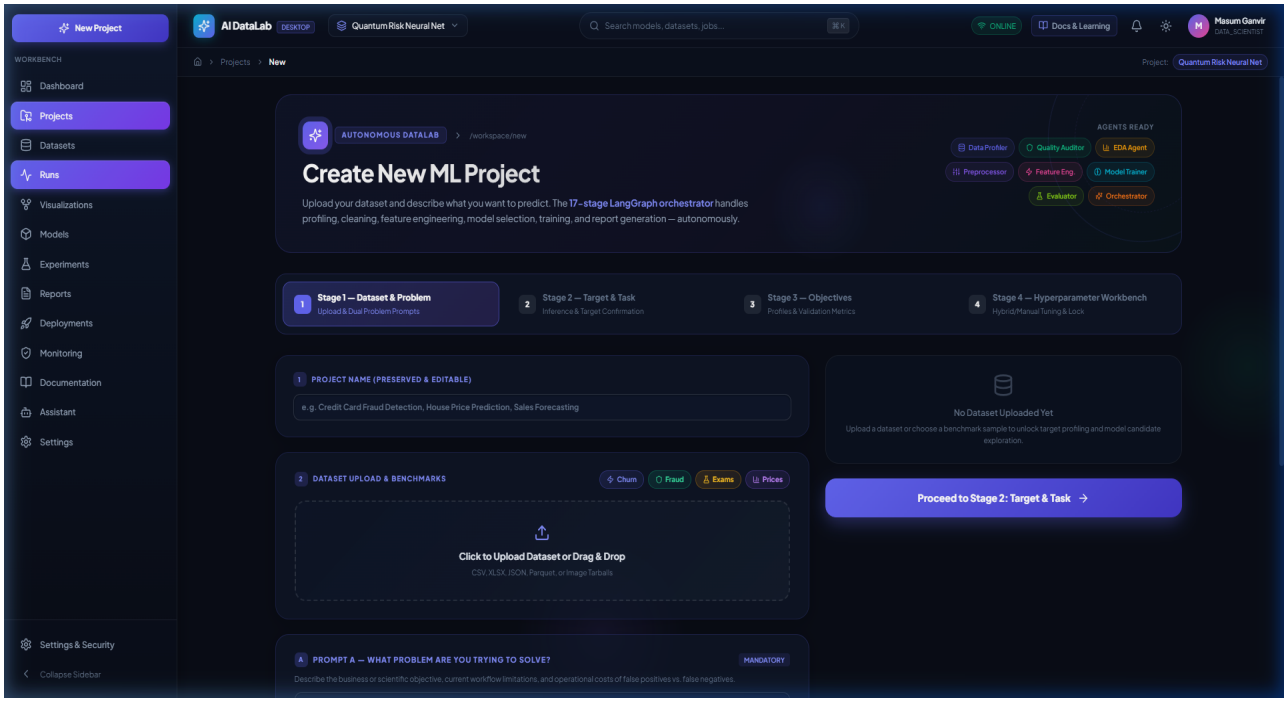


Figure: Step 2: Defining Intent through the Two-Prompt System

3. The Two-Prompt Problem Formulation System

DataWise AI uses a structured two-prompt dialog to ensure the AI understands your domain:

- **Prompt 1: What problem are you trying to solve?** Example: *'Predict customer churn for a telecom subscription service to reduce monthly cancellations.'*
- **Prompt 2: What solution do you want, and how should it work?** Example: *'A binary classification model that emphasizes recall on churners, trains in under 5 minutes, and outputs a clean scikit-learn pipeline.'*

4. The Interactive Data Science Workbench

Once intent is established, DataWise AI opens the unified Workbench. Here, you interact with the autonomous agent team. The agents analyze your dataset, detect outliers, propose imputations, and request your approval before applying changes.

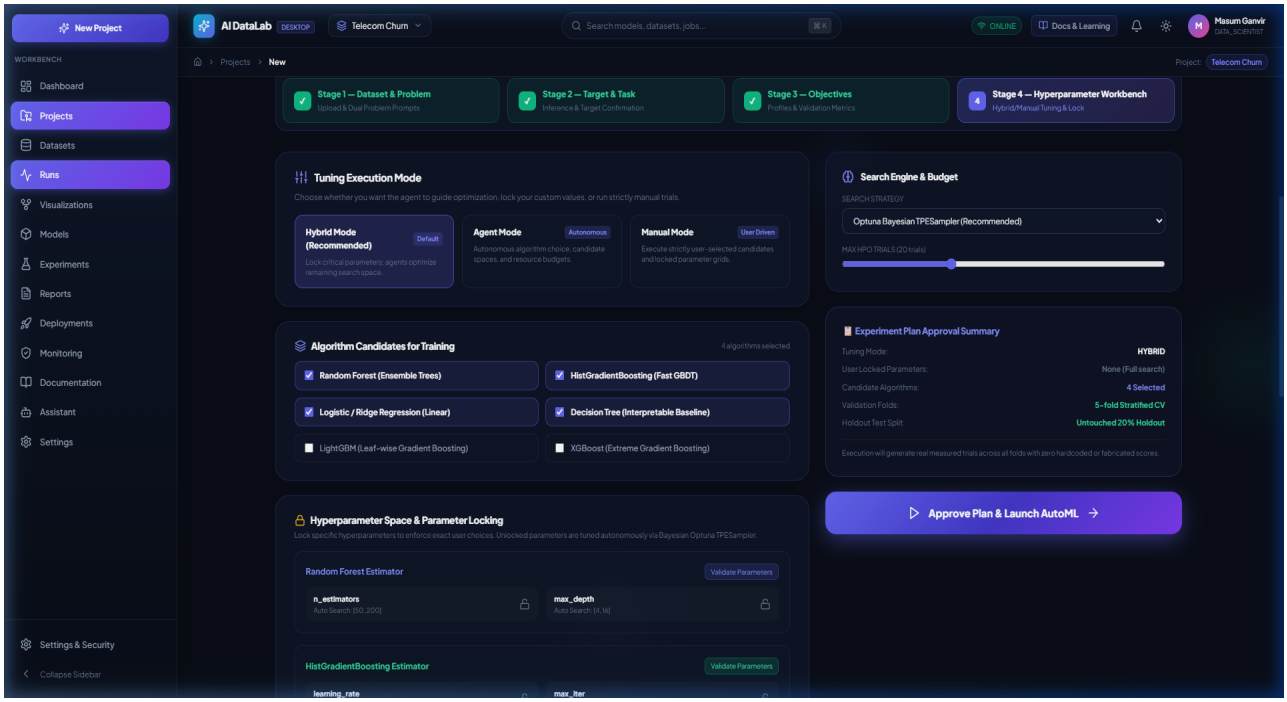


Figure: Step 3: Interactive Workbench, EDA Visualizations, and Agent Chat

5. Hyperparameter Customization & Model Leaderboard

In the Hyperparameter tab, you can inspect candidate algorithms selected for your task (e.g., XGBoost, LightGBM, Random Forest). You can choose between Fast Automated Search, Balanced Tuning, or Deep Optuna Bayesian Optimization. Once executed, the model leaderboard displays validation scores (F1, ROC-AUC, Accuracy, RMSE) alongside inference latency.

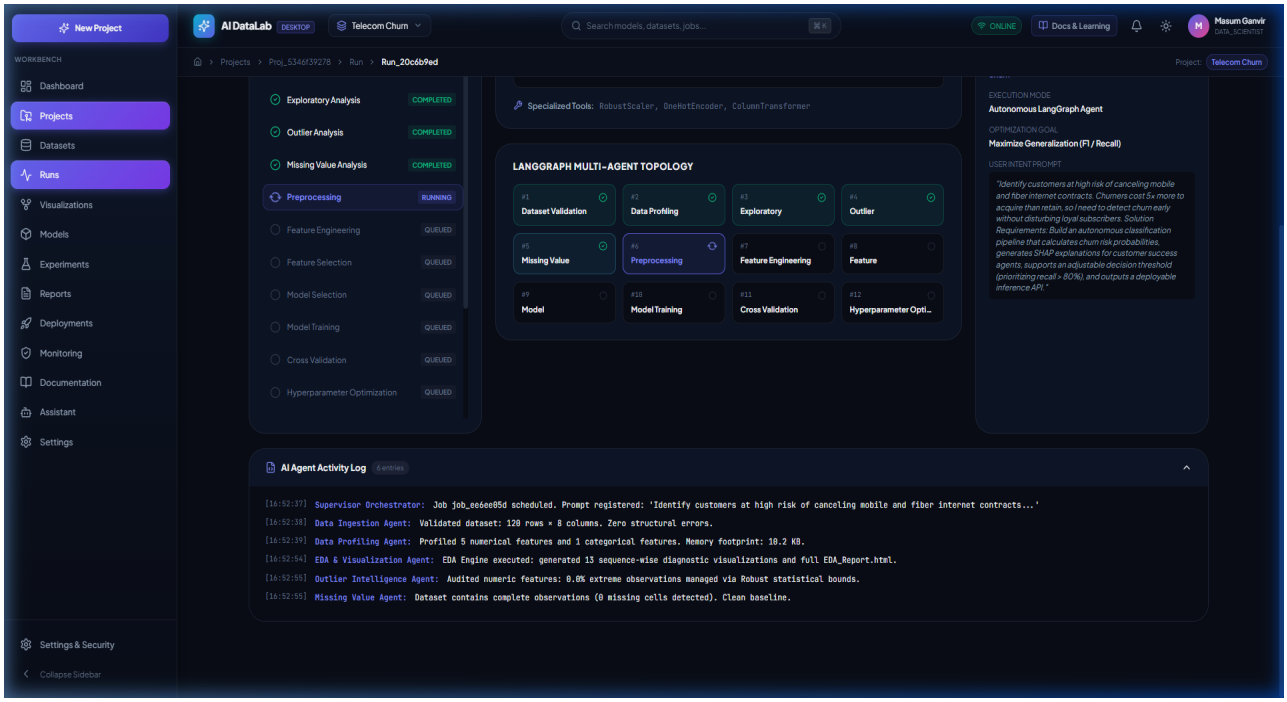


Figure: Step 4: Real-Time Experiment Execution & Model Training Stream

6. Exporting Production Artifacts

When you select your preferred model, click **'Export Artifacts'**. You receive a downloadable bundle containing:

- `model_pipeline.pkl` — The fully fitted preprocessing and model pipeline ready for instant inference.
- `analysis_report.pdf` — Executive summary of data quality, feature importances, and validation curves.
- `reproducible_workflow.ipynb` — Standalone Jupyter notebook containing all code to recreate the model.
- `serve_api.py` — A production FastAPI script to deploy the model as a microservice.